

Miller Alejandro Ospina Londoño

Full Stack Developer | Red Teaming Specialist | **Backend & Security Engineer** 📍 Chía, Cundinamarca | 📞 (+57) 313-253-3008 | ✉️ analyticshydra@gmail.com
Languages: English (C1 - Advanced), French (Medium), Spanish (Native)

Professional Profile:

PROFESSIONAL PROFILE

Full Stack Developer and Red Teaming Specialist with hands-on experience delivering **production-grade, security-first applications** in distributed and cloud-based environments. Specialized in designing and implementing **Zero-Trust architectures**, high-performance backend systems, and **concurrency-safe solutions** using **Django, Redis, and C**.

Possesses strong expertise in **low-level system programming and Windows API**, enabling deep visibility into operating system behavior, attack surfaces, and defensive engineering strategies. Experienced in developing and analyzing security tooling, as well as building resilient platforms capable of **detecting, resisting, and mitigating advanced attack vectors**.

Brings a systems-oriented engineering mindset combined with proven technical leadership, effectively aligning **backend architecture, security requirements, and business objectives**. Demonstrated ability to deliver **scalable, fraud-resistant solutions** in cloud environments such as **AWS and Railway**, while communicating complex security concepts clearly to cross-functional and international teams.

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💻 KEY PROJECTS

Secure Event Ticketing Platform ("Ceviche" Architecture) | *Full Stack Architect*

*Designed a high-performance, **Zero-Trust** ticketing platform focusing on concurrency and anti-fraud mechanisms.*

- **Architecture:** Implemented a **Hexagonal & Event-Driven Architecture** using Python/Django to isolate business logic from external adapters (Wompi, SIIGO).
- **Concurrency Control:** Solved the "double-booking" problem using **Redis + Lua Scripting** for atomic seat locking transactions.
- **Security:** Developed a **Finite State Machine** to manage QR lifecycles (Entry/Exit/Re-entry) preventing replay attacks. Implemented Field-Level Encryption (Fernet) for sensitive user data.
- **Resilience:** Integrated **Cloudinary** for static media persistence and **Celery** for asynchronous billing tasks with exponential backoff strategies.

Hell's Gateway Project | *Red Teaming & System Programming*

Developed a custom C-based artifact demonstrating low-level Windows OS interaction for security research purposes.

- **Windows API:** Leveraged native APIs (`CreateFileA`, `WriteFile`, `CloseHandle`) to manipulate the file system directly, bypassing high-level abstractions.
- **Implementation:** Created a robust executable capable of verifying file existence, writing custom payloads, and managing system handles efficiently.
- **Compilation:** Optimized build process using **MinGW** and **GCC** for lightweight executable generation.

Automated LAN Vulnerability Analyzer | *Cybersecurity Tooling*

- Built an automated Python script to scan local networks, identifying open ports and potential security gaps.
- Applied mathematical logic to optimize scanning speeds and reporting accuracy.

Learning Labs & Mirror Code | *EdTech Development*

- Created interactive learning environments and simulation code to teach complex programming concepts to students, bridging the gap between theoretical physics and applied coding.

Education

● IED Jose Joaquin Casas, Chía Cundinamarca
● Universidad Agraria de Colombia, Mechatronics Engineering 4 semesters
●Sena Software Development
● BackTrack Academy, CPE Certified ●Pentester Engineering
● Python Code Academy, Neural Networks.
● CCNA Cisco
● Python Cisco
● Vex Coach, REC Foundation coach
● CEH Cisco

💼 PROFESSIONAL EXPERIENCE

Teacher in Robotics, Physics & AI | *Gimnasio Britanico IB School Jan 2024 - May 2025*

- Led the implementation of **Python and AI curriculums**, teaching students to integrate neural networks with physical robotics systems.
- Coached robotics teams using VEX technology, applying engineering principles to competitive environments.

Mathematics & Physics Lead / Onboarding Specialist | *Various Institutions & Sitel 2016 - 2023*

- **Sitel (Cox Communications):** Managed onboarding processes for technical teams, ensuring 100% adherence to corporate protocols and technical requirements.
- **UWI School / El Cafetal:** Head of Department. Revolutionized the curriculum by introducing **Reverse Engineering** methodologies to teach Physics and

Mathematics, significantly improving student learning curves.

Githubs

https://github.com/OMiller555/Hell-s-Gateway-Handler.git

https://github.com/OMiller555/labslearning2.git

https://github.com/Hydra0101/clonemirror.git

ALL OF THEM ARE MINE